

MATH 4790 Midterm Spring 2026

Name: Key Auburn ID: _____

Instructions

1. Print your full name and your banner ID on your exam. Then finish reading these instructions, and sign the bottom of this page.
2. All electronic equipment (cell phones, Apple Watch, AirPod, etc) are **NOT ALLOWED** during the exam time.
3. You are allowed to use up to 2 calculators from the following list: BA-35, BA II Plus, BA II Plus Professional, TI-30Xa, TI-30X II (IIS solar or IIB battery), TI-30XS MultiView (or XB battery).
4. No talking is allowed during the exam.
5. Unless otherwise indicated, **SHOW ALL YOUR WORK**. If no work is shown, no partial credit can be awarded. Your work and answer need to be accurate and relevant to receive points.
6. You will be given exactly 50 minutes for this exam.
7. Any student not following the above instructions nor behaving according to the above instructions during the exam may have their exam confiscated and points deducted.

I have read and fully understand all of the above instructions.

Signature: _____

Do not write in the table below

Problem	Points	Score
1	11	
2	11	
3	11	
4	11	
5	11	
6	11	
7	11	
8	11	
9	11	
Total	99+1	

Problem 1. Ace-My-Test Bank has liabilities of \$4,000 due in one year, \$3,000 due in two years, and \$2,000 due in three years. The bank may invest in the following bonds:

- **Bond A:** A 1-year zero-coupon bond with annual effective yield i .
- **Bond B:** A 2-year bond with annual coupons at 5% and yield 4% effective annually.
- **Bond C:** A 3-year bond with annual coupons at 6% and yield 6% effective annually.

Suppose the bank constructs a cash flow matching portfolio using these bonds. If the total amount invested in Bond A is \$3,567.39, determine i .

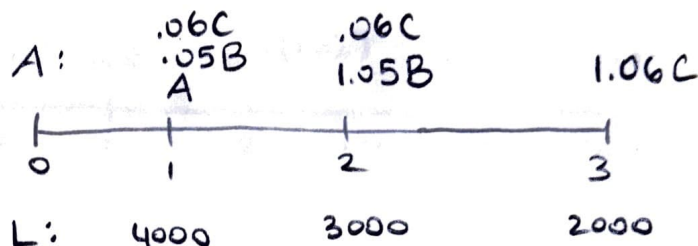
A. 4.8%

B. 5.1%

C. 5.4%

D. 5.8%

E. 6.0%



$$2000 = 1.06C \Rightarrow C = 1886.79$$

$$3000 = .06C + 1.05B \Rightarrow B = 2749.33$$

$$4000 = .06C + .05B + A \Rightarrow A = 3749.33$$

$$A(1+i)^{-1} = 3567.39 \Rightarrow i = 5.1\%$$

Problem 2. The annual effective spot rate for maturity t (in years) is given by the function

$$s_t = \begin{cases} 0.04, & 1 \leq t \leq 10, \\ 0.06, & 11 \leq t \leq 20. \end{cases}$$

A 20-year annuity-immediate pays \$100 at the end of each year and each subsequent payment increase by 2%. Find its present value.

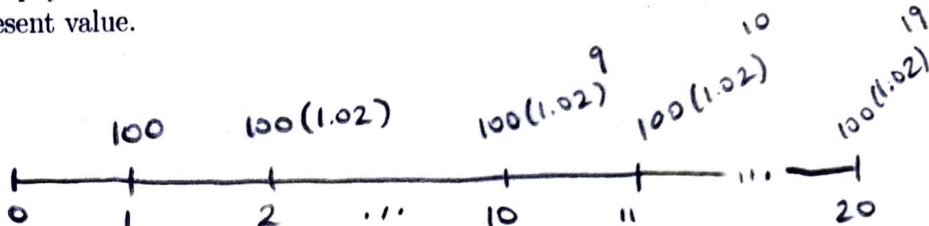
A. 1,389

B. 1,401

C. 1,426

D. 1,511

E. 1,589



$$\begin{aligned} PV &= 100(1+s_1)^{-1} + 100(1.02)(1+s_2)^{-2} + \dots + 100(1.02)^9(1+s_{10})^{-10} \\ &\quad + 100(1.02)^{10}(1+s_{11})^{-11} + 100(1.02)^{11}(1+s_{12})^{-12} + \dots + 100(1.02)^{19}(1+s_{20})^{-20} \\ &= 100(1.04)^{-1} + 100(1.02)(1.04)^{-2} + \dots + 100(1.02)^9(1.04)^{-10} \\ &\quad + 100(1.02)^{10}(1.06)^{-11} + 100(1.02)^{11}(1.06)^{-12} + \dots + 100(1.02)^{19}(1.06)^{-20} \\ &= 100 \left[\frac{1 - \left(\frac{1.02}{1.04}\right)^{10}}{.04 - .02} \right] + 100(1.02)^{10}(1.06)^{-10} \left[\frac{1 - \left(\frac{1.02}{1.06}\right)^{10}}{.06 - .02} \right] \end{aligned}$$

$$= 1425.84$$

Problem 3. You are given that the force of interest at time t (in years) is

$$\delta_t = \frac{1}{2t+5}.$$

Find the effective interest rate in the 5th year.

A. 4.4%

B. 5.4%

C. 6.4%

D. 7.4%

E. 8.4%

$$i_{[4,5]} = \frac{a(5) - a(4)}{a(4)}$$

$$a(t) = e^{\int_0^t \frac{1}{2r+5} dr} = e^{\frac{1}{2} \ln|2r+5| \Big|_0^t}$$

$$= e^{\frac{1}{2} (\ln(2t+5) - \ln(5))}$$

$$= \frac{e^{\ln(2t+5)^{\frac{1}{2}}}}{e^{\ln(5)^{\frac{1}{2}}}} = \left(\frac{2t+5}{5} \right)^{\frac{1}{2}} = \sqrt{\frac{2t}{5} + 1}$$

$$i_{[4,5]} = \frac{\sqrt{3} - \sqrt{2.6}}{\sqrt{2.6}} = 7.41\%$$

Problem 4. Maya purchases a portfolio of two zero-coupon bonds.

- Bond 1 matures in 4 years for \$327.70
- Bond 2 matures in 8 years for \$1,288.64.

The annual effective yield is 7.0%. Calculate the modified convexity of Maya's portfolio.

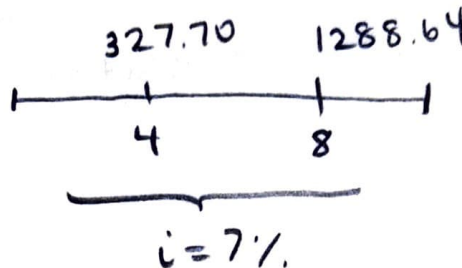
A. 40.18

B. 43.92

C. 45.98

D. 50.08

E. 51.53



Let $P = \%$ of the portfolio's PV in Bond 1

$$P = \frac{327.70(1.07)^{-4}}{327.70(1.07)^{-4} + 1288.64(1.07)^{-8}} = \frac{250}{1000} = \frac{1}{4}$$

$$MacD = 4\left(\frac{1}{4}\right) + 8\left(\frac{3}{4}\right) = 7$$

$$MacC = 16\left(\frac{1}{4}\right) + 64\left(\frac{3}{4}\right) = 52$$

$$ModC = (1.07)^{-2} [7 + 52] = 51.53$$

Problem 5. ~~Jerry and Sally both~~

Jerry invests in a perpetuity which yields payments at the end of each year starting with P , and each subsequent payment increases by \$10.

Sally invests in an annuity-due which yields payments at the beginning of the year 10 years. The first payment is \$1,000 and each subsequent payment increases by 3%.

The annual effective interest rate is 5%.

Both investments have the same price. Find P .

A. 245.75

B. 259.25

C. 271.80

D. 284.50

E. 296.10

$$I_{P,10} a_{\infty|5\%} = 1000 (1.05) \left[\frac{1 - \left(\frac{1.03}{1.05} \right)^{10}}{.05 - .03} \right]$$

||

$$\frac{P}{.05} + \frac{10}{.05^2}$$

$$P = 259.25$$

Problem 6. Bob's portfolio is a level perpetuity-immediate valued at an annual effective yield of i .

Tina's portfolio consists of two cash flows: \$100 at time $t = 5$ and \$200 at time $t = 25$, valued at an annual effective yield of 6%.

If the modified duration of Bob's portfolio equals the modified duration of Tina's portfolio, find i .

A. 7.50%

B. 8.36%

C. 9.10%

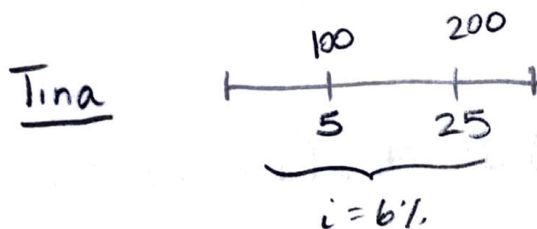
D. 10.00%

E. 11.20%

$$\text{Mod } D(\text{Bob}) = \text{Mod } D(\text{Tina})$$

Bob

$$P(i) = \frac{1}{i} \Rightarrow P'(i) = -\frac{1}{i^2} \Rightarrow \frac{-P'(i)}{P(i)} = \frac{1}{i} \Rightarrow \text{Mod } D(\text{Bob}) = \frac{1}{i}$$



$$\begin{aligned} \text{Mac } D(\text{Tina}) &= \frac{100(5)(1.06)^{-5} + 200(25)(1.06)^{-25}}{100(1.06)^{-5} + 200(1.06)^{-25}} \\ &= 12.6818 \end{aligned}$$

$$\begin{aligned} \text{Mod } D(\text{Tina}) &= \text{Mac } D(\text{Tina}) (1.06)^{-1} \\ &= 11.9639 \end{aligned}$$

$$\frac{1}{i} = 11.9631 \Rightarrow i = 8.358\%$$

Problem 7. Carlos has a liability of \$11,000 due in 8 years. This liability will be met with payments of \$4,500 in 5 years and B in b years for some $b > 8$.

Carlos is employing a full immunization strategy using an annual effective interest rate of 17%.

Calculate B .

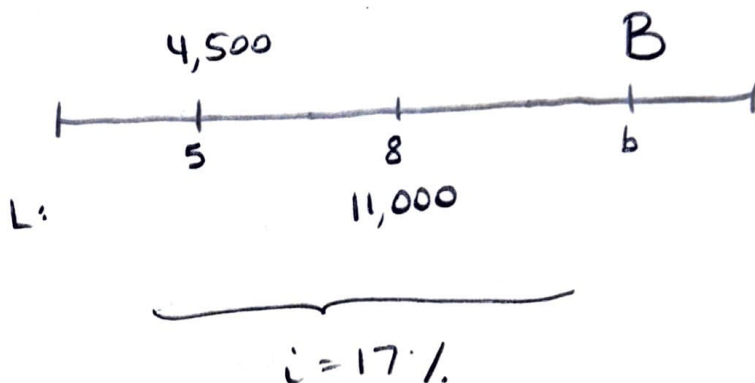
A. 8,987

B. 9,045

C. 9,283

D. 10,023

E. 10,460



Let P be the % of the asset portfolio in 5-year investment

$$P = \frac{4500(1.17)^{-5}}{PV(\text{Assets})} = \frac{4500(1.17)^{-5}}{PV(\text{Liabilities})} = \frac{4500(1.17)^{-5}}{11,000(1.17)^{-8}} = .6552$$

$$MacD(\text{Assets}) = MacD(\text{Liabilities})$$

$$5P + b(1-P) = 8 \Rightarrow b = 13.7$$

$$PV(\text{Assets}) = PV(\text{Liabilities})$$

$$4500(1.17)^{-5} + B(1.17)^{-13.7} = 11,000(1.17)^{-8}$$

$$B = 9281.38 \leftarrow \text{slightly off by rounding } b \text{ to } 13.7$$

Problem 8.

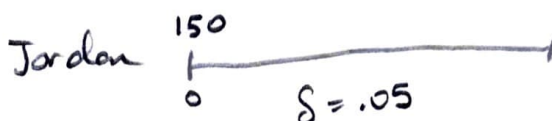
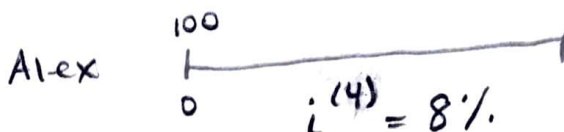
Alex deposits \$100 into Fund A, which earns a nominal annual interest rate of 8% convertible quarterly.

At the same time, Jordan deposits \$150 into Fund B, which earns a constant annual force of interest of 0.05.

After n years, the balance of Fund A equals the balance of Fund B.

Calculate n .

- A. 9.4
- B. 10.6
- C. 11.8
- D. 12.7
- E. 13.9



$$100 \left(1 + \frac{.08}{4}\right)^{4n} = 150 e^{.05n}$$

$$\left((1.02)^4\right)^n = 1.5 (e^{.05})^n$$

$$(1.02)^4 = 1.5^{1/n} e^{.05}$$

$$\frac{1.02^4}{e^{.05}} = 1.5^{1/n}$$

$$\ln\left(\frac{1.02^4}{e^{.05}}\right) = \frac{1}{n} \ln(1.5) \Rightarrow n = \frac{\ln(1.5)}{\ln\left(\frac{1.02^4}{e^{.05}}\right)}$$

$$\approx 13.8808$$

Problem 9. A continuous annuity pays at a rate of $3e^{.08t}$ where t is in years. The fund earns interest at a constant force of interest $\delta = 0.05$.

Find the accumulated value at time 10 of this annuity.

A. 54.13

B. 55.99

C. 57.68

D. 59.42

E. 61.27

$$f(t) = 3e^{.08t}$$

$$\delta = .05 \leftarrow \text{compound interest}$$

$$PV = \int_0^{10} 3e^{.08t} \cdot e^{-.05t} dt$$

$$= 3 \int_0^{10} e^{.03t} dt = \frac{3}{.03} e^{.03t} \Big|_0^{10} = 100(e^{.3} - 1)$$

$$\begin{aligned} FV &= PV \cdot a(10) = 100(e^{.3} - 1)e^{.05(10)} \\ &= 100(e^{.8} - e^{.5}) \\ &= 57.682 \end{aligned}$$